

Benefits of data partitioning

Data partitioning is the process of dividing data into separate data segments, or partitions that can be managed and accessed separately. Partitions create smaller sections of your dataset, which can then be queried and analyzed individually. In this reading, you'll review types and uses of partitioning, and learn some of the benefits of partitioning data.

Data partitioning basics

At a high level, a database is a set of schema, or datasets. Within each schema, there are many tables. There are several methods of partitioning data tables. Each provides a different format for data, and different options for querying. Depending on the needs of your organization, there are several scenarios where partitioning may be useful.

Reasons to partition data

Data partitioning is a data management technique that creates subsets of data that are easier to manage. Some reasons to partition databases include:

- Dividing data by usage
- Dividing data by subgroups, like regions or users
- Maintaining smaller storage sizes
- Archiving historical data
- Distributing data across multiple computers

Benefits of data partitioning

Partitioning data tables may give you a manageable amount of data to work with, and can be divided based on your specific needs. There are additional benefits to partitioning, including:

- Increasing query speeds
- Developing scalability parameters
- Reducing the cost of storage
- Improving security
- Increasing availability

Working with smaller datasets also gives you the ability to review and enhance the datasets on a smaller scale, increasing your ability to ensure accuracy.

Increase query speeds

For example, consider an agricultural company that stores data on monthly crop yield, and monthly sales. Since the crops change every season, they have partitioned their data based on seasons. This means that when searching for the total sales of a single crop during the spring, the corporate headquarters can query only the spring database and sales table. The smaller dataset means that data can be queried quickly.

Develop scalability parameters

Now that the agricultural company has divided its data into partitions, it has created ingestion parameters for each partition. As new data is generated, it's ingested into the partitions. This allows for scalability of the datasets since the tables have already been defined.

Reduce storage costs

After collecting data for five years, the agricultural company partitions its datasets again. This time, they partition the data by year. Datasets three years or older are hosted in storage resources that are slower, but cheaper. Since the organization does not need to access this data regularly, it saves on storage costs.

Improve security

The agricultural company partitions its datasets again. This time, it separates data that includes information about their own unique hybrids, and schedules it into a separate partition. The data about annual crop production is kept in a partition that's available to share publicly.

Increase availability

As the organization grows, and farmland is acquired across the country, there is an added need for consistent uptime in various time and availability zones. The company now has six to seven teams around the world that require consistent access to data. With partitioned datasets, the agricultural company is able to create increased availability since there's not a single point of failure for the dataset.

Key takeaways

Database partitioning allows data teams to divide their databases into manageable datasets. In larger organizations, partitioned databases increase availability, improve security, save on storage costs, and increase the speed and efficiency of data querying. Database partitioning also provides increased performance, and can be an effective method of database management.

Resources for more information

To learn more about how to partition databases, and the benefits of partitioning, check out this link:

- Google Cloud Community provides information about how database partitions work:
<https://www.googlecloudcommunity.com/gc/Modeling/Database-Partitions-How-they-work/m-p/578402>